METRIC
DOD-P-24631/6(SH)
13 April 1984

MILITARY SPECIFICATION SHEET

PAINT, CHLOROSULFONATED, POLYETHYLENE, WHITE, NAVY FORMULA 189, TYPE III (METRIC)

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the compound described herein shall consist of this specification and the latest issue of DOD-P-24631(SH).

Formula: This formula covers white chlorosulfonated polyethylene paint designated Navy Formula 189 for exterior use. The paint shall consist of the ingredients specified in the quantities specified.

Component A		Pounds	Kilograms	Gallons[1/]	Liters[1/]
Chlorosulfonated polyethylene	Hypalon 20, DuPont	618.3	280.6	67.36	255.0
Trimal	Tribasic Lead Maleate, Assoc. Lead, Inc.	247.2	112.1	4.70	17.8
Titanium dioxide	ASTM D 476, type III or IV	309.2	140.3	9.26	35.0
Aromatic solvent	HI SOL 10, Ashland	164.4	74.6	22.51	85.2
Ketone	Methyl Amyl Ketone	383.6	174.0	56.26	212.9
n-Butyl acetate	TT-B-838	1918.4	870.2	260.35	985.4
Ethylene glycol mono ethyl	TT-E-781	274.1	124.3	35.28	133.5
Thickener	M-P-A-60X, NL Ind.	28.9	13.1	3.98	15.1
Dispersant	Tenlo 70, Diamond Shamrock	2.2	1.0	0.27	1.0
		3946.3	1790.2	459.97	1740.9

1/ Provided for guidance only - Actual values are dependent on variation in raw meterials.

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	Component B	Pounds	Kilograms	Gallons[1/]	Liters[1/]
Wood rosin	Staybelite resin 10, Hercules	29.29	5.5	3.42	12.9
Curing agent	NA22, DuPont	6.18	2.8	0.52	2.0
Curing agent	DOTG, American Cyanamid	12.37	5.6	1.24	4.7
Aromatic solvent	HI SOL 10, Ashland	2.06	0.9	0.28	1.1
Ketone	Methyl Amyl Ketone	4.79	2.2	0.70	2.6
n-Butyl acetate	TT-B-838	24.00	10.9	3.26	12.3
Ethylene glycol mono ethyl ether	TT-E-781	3.47	1.6	0.45	1.7
		82.16	29.5	9.87	37.3

1/ Provided for guidance only - Actual values are dependent on variation in raw materials.

NOTES:

1. The chlorosulfonated polyethylene shall be "Hypalon 20" available from DuPont, Wilmington, DE, or equivalent (see figure 1).
2. The carbon black shall be "Raven 8000 beads" available from Columbian Carbon, Akron, OH, or equivalent.
3. The thickener shall be "M-P-A-60X" available from N.L. Industries, Inc., Hightstown, NJ, or equivalent.
4. The dispersant shall be "Tenlo 70" available from Diamond Shamrock, Morristown, NJ, or equivalent.

Use of alternative ingredients in this formula must have prior approval by the Naval Sea Systems Command. Approval will be based on review of data showing equivalent physical and chemical characteristics to the specified ingredient. It will be necessary to demonstrate that the paint compound made using the alternative ingredient will conform to all requirements of this specification.

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Quantitative requirements

Requirements	Component A		Component B		Mixed component	
	Min	Max	Min	Max	Min	Max
Pigment: Percent by mass of paint	12.2	16.2	0	0		
Percent by volume of paint	1.2	6.2	0	0		
Volatiles: Percent by mass of paint	68.0	72.0	39.6	43.6		
Percent by volume of paint	80.0	84.0	45.6	49.6		
Nonvolatiles: Percent by mass of paint	13.8	17.8	56.4	60.4		
Percent by volume of paint	12.8	16.8	50.4	54.4		
Consistency, Krebs units					81.0	
Mass per unit volume: lbs/gallon	8.33	8.83	8.11	8.61	8.31	8.82
Kg/L	1.01	1.06	0.98	1.02	1.01	1.06
Fineness of grind (Hegman)					6.0	---
Flash point, Pensky-Martens °C (°F)				18.0	(65.0)	---
Wet reflectance color (R_y), percent					---	---
Set-to-touch time: hours at 4.4°C (40°F)					---	3.0
hours at 23°C (73°F)					---	2.0
Dry-hard hours at 4.4°C (40°F)					---	3.0
hours at 23°C (73°F)					---	2.0
Titanium dioxide, percent of pigment	7.89		0			
Pot life, hours at 23°C (73°F)					8.0	---
Gloss, 16°C (60°F) specular, percent					---	---
Contrast ratio					0.9	---

Preparing activity:
Navy - SH
(Project 8010-N251)

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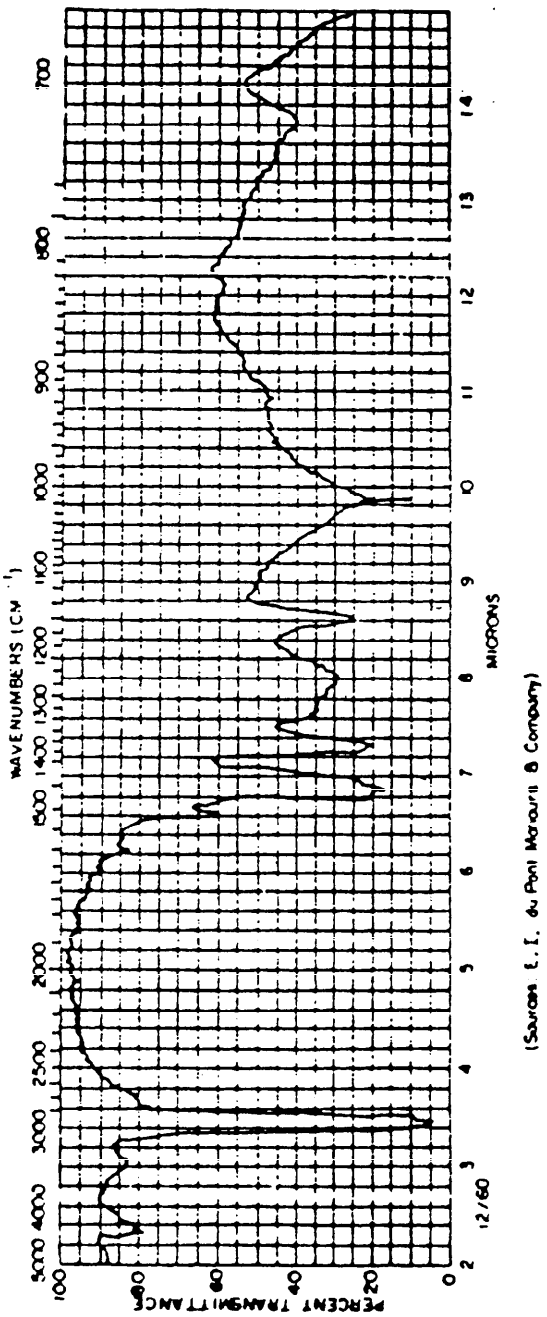


FIGURE 1. Infrared spectrogram of Hypalon 20 (chlorosulfonated polyethylene).

SH 12147

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